14 JULY 2021

Meet Johan.

11 · Bad Neuenahr-Ahrweiler

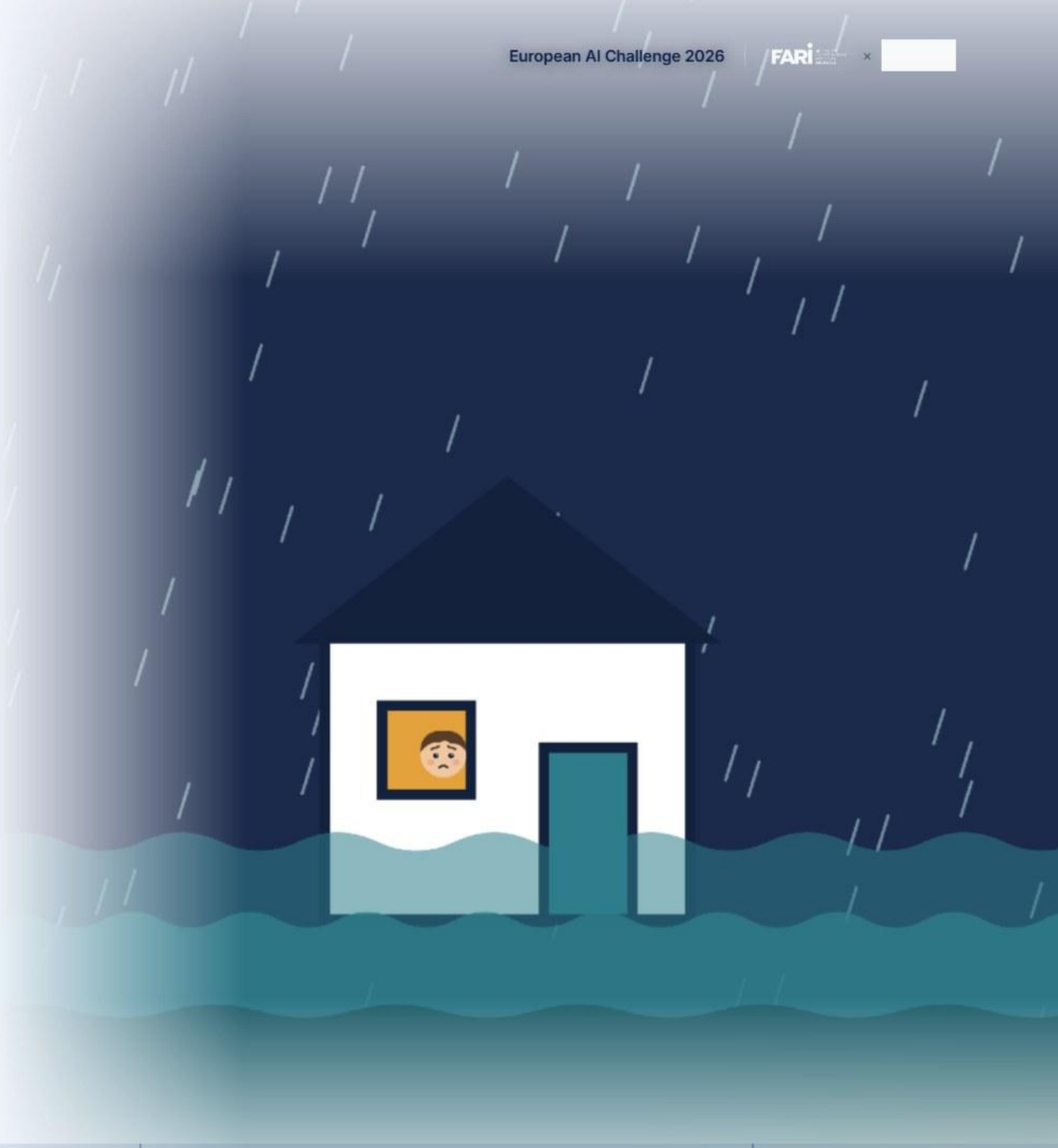
— Fictional character



The water came at night.

85% of those warned didn’t expect severe flooding

46% didn’t know what to do



Thieken et al. 2023, Natural Hazards and Earth System Sciences 23, 973-990 · online survey, n = 1,315, North Rhine-Westphalia and Rhineland-Palatinate

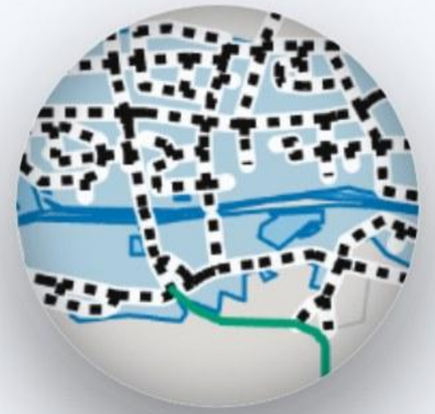
01 **Am I in danger?**

02 **How serious is it?**

03 **Where should I go?**



From flood map to one clear message.



01

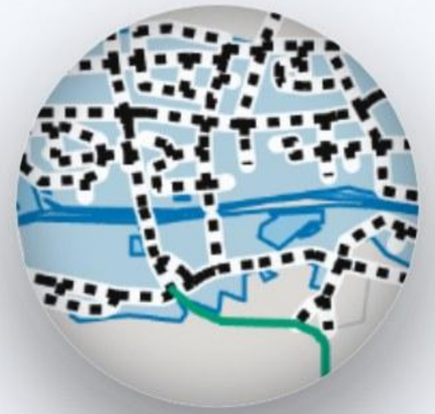
Flood + roads

→ closed roads

Copernicus EMS · EFAS · OpenStreetMap

Supports, and does not replace, official warnings and evacuation decisions.

From flood map to one clear message.



01

Flood + roads

→ closed roads

Copernicus EMS · EFAS · OpenStreetMap



02

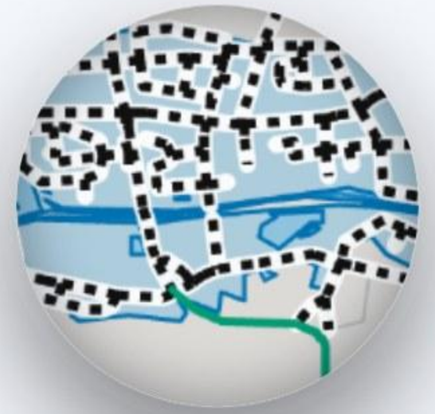
Match + route

→ a shelter with room

GHS-POP · building capacity · open roads

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01

Flood + roads

→ closed roads

Copernicus EMS · EFAS · OpenStreetMap



02

Match + route

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GHS-POP · building capacity · open roads



03

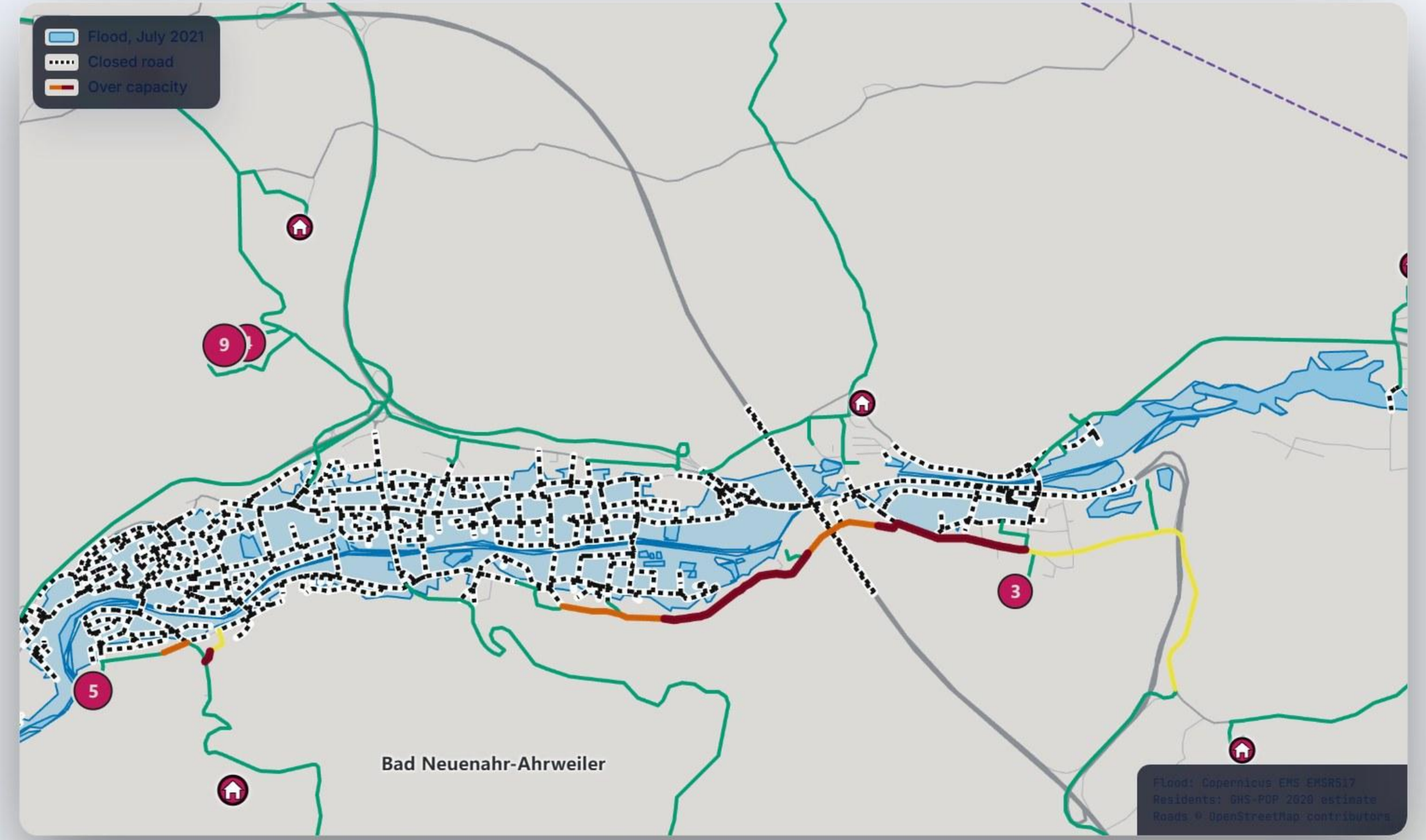
Johan's alert

→ three answers, plain text

Cell Broadcast · works on 2G

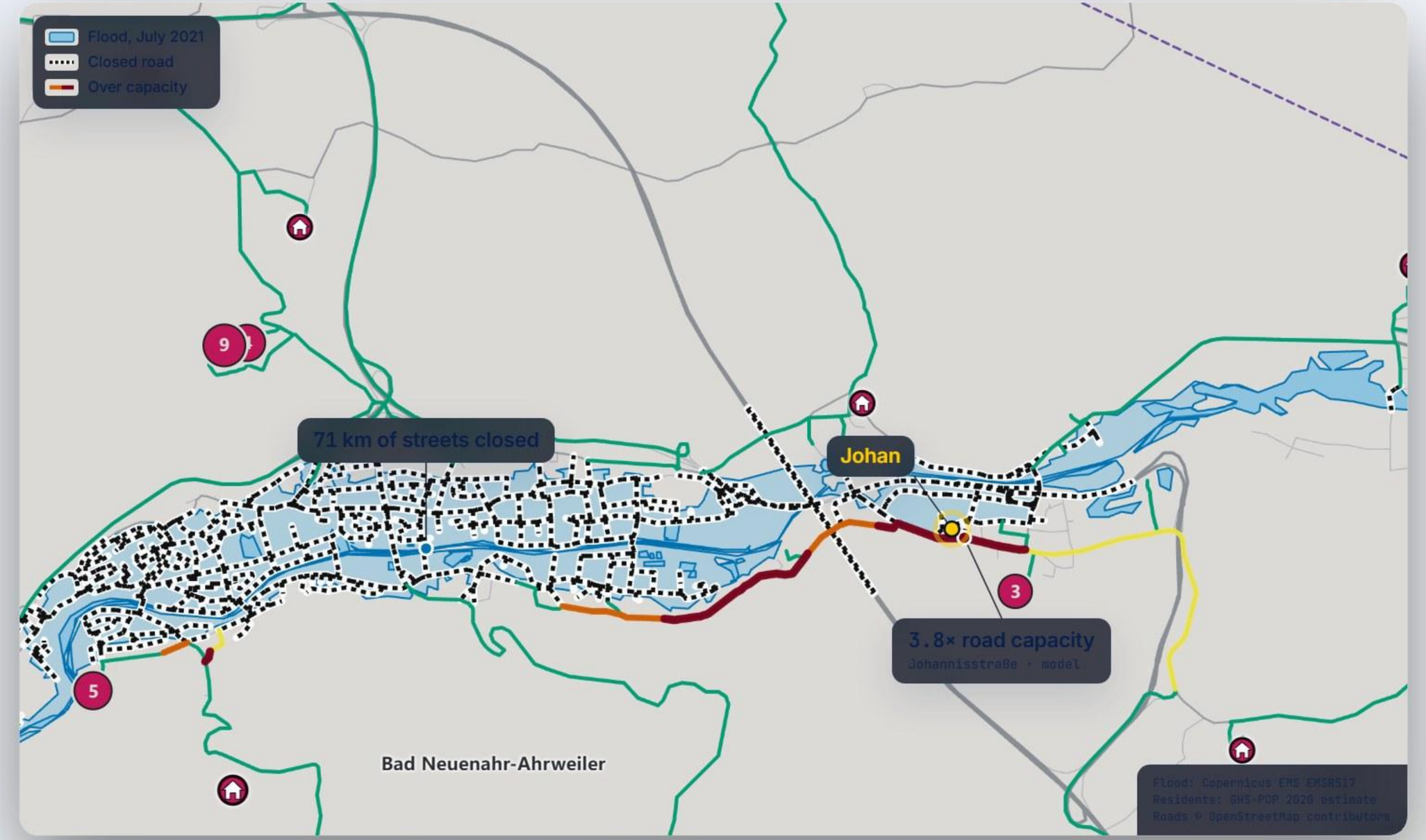
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Streets close. Exits overload.



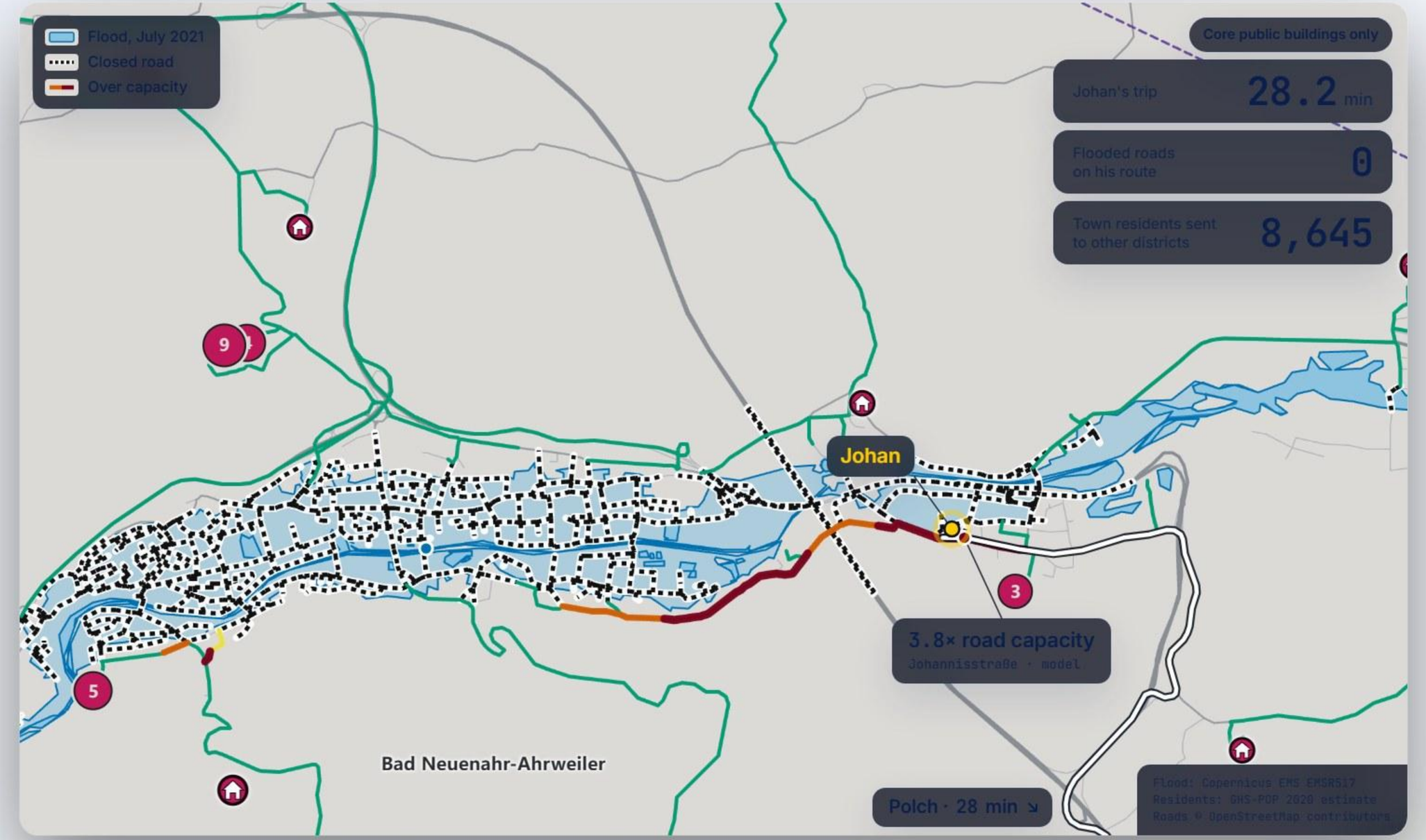
Smart Shelter System · bundle 2026-10-30 09:46 UTC (commit 01abbb0) · capacity, car share and speeds are assumptions; stress is an indicator, not a forecast

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Smart Shelter System · bundle 2026-10-30 09:46 UTC (commit 01abbb0) · capacity, car share and speeds are assumptions; stress is an indicator, not a forecast

Decide before the night.

FLOOD LEVEL

2021 (MAPPED) · 10-yr · 100-yr · 500-yr (MODEL)

SHELTER BUILDINGS

Core public · + Extended

78% reach a shelter within 30 min



within 30 min **19,920** · over 30 min **5,639** · no place **0**

ROADS OVER CAPACITY **37** · MEAN TRIP **22.7** MIN · RESIDENTS **25,559**

Kreis Euskirchen + Landkreis Ahrweiler · residents inside the flood: GHS-POP 2020 estimate · buildings: OpenStreetMap candidates, not designated shelters · Core public = set A, 5 m² per person; + Extended = set B, 3.5 m² per person (assumptions) · Smart Shelter System bundle 2026-10-30

Decide before the night.

FLOOD LEVEL

2021 (MAPPED) · 10-yr · 100-yr · 500-yr (MODEL)

SHELTER BUILDINGS

Core public · + Extended

78% reach a shelter within 30 min

28.2 min

walk 1.3 · car 27.0

Polch



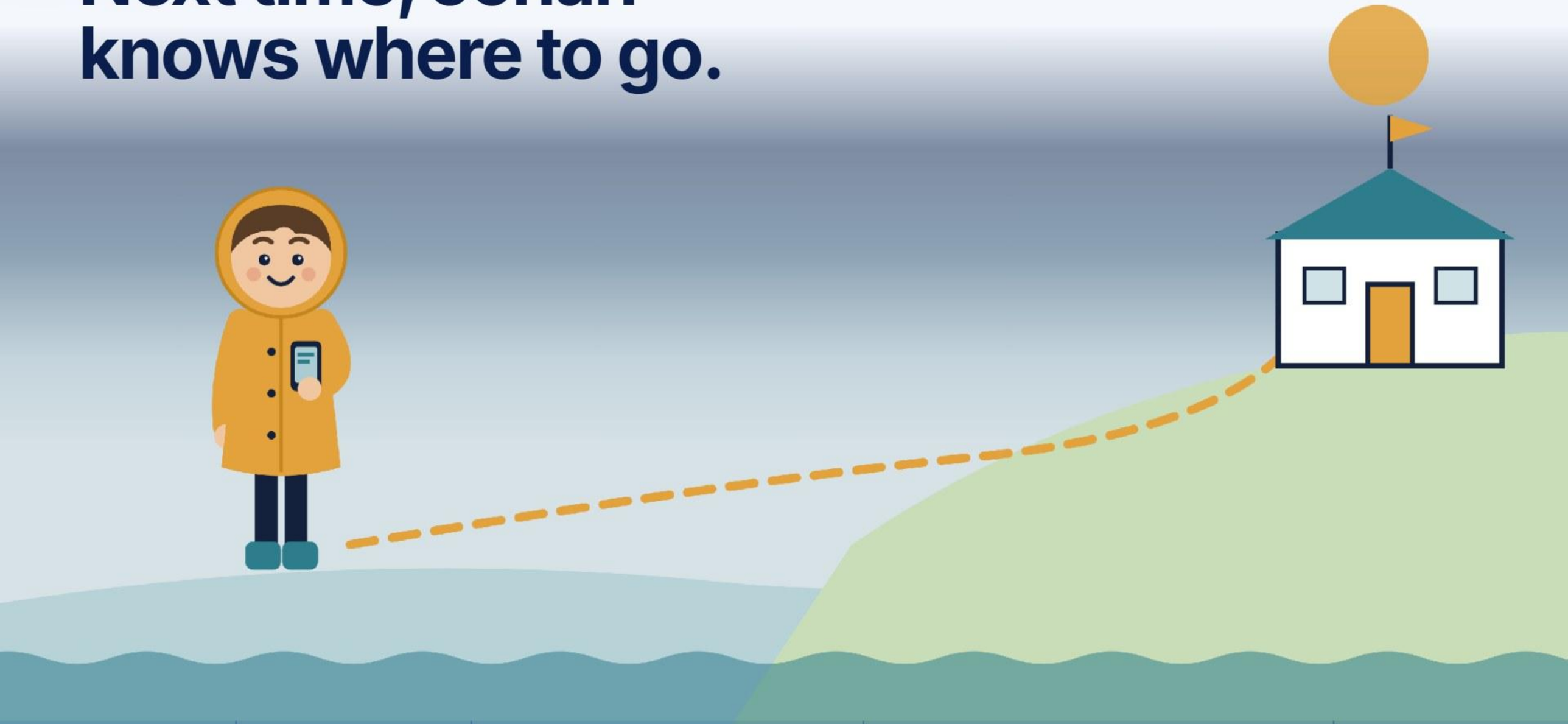
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Next time, Johan knows where to go.



~230

lives lost

€32–53 bn

damage

~10×

return on early warning

Built only on open EU-wide data:

Copernicus EMS · EFAS · GHSL · OpenStreetMap



July 2021 floods in BE, DE and NL. Deaths and damage: team policy brief refs [2], [9], [20]; return on early warning: WMO 2022, ref [24]

TO CRISIS AUTHORITIES IN BE · DE · NL

1. **Designate and publish flood shelters.**
2. **Make every alert answer three questions.**
3. **Drill the route at school.**



TEAM 1

European AI Challenge 2026

FARI × [redacted]